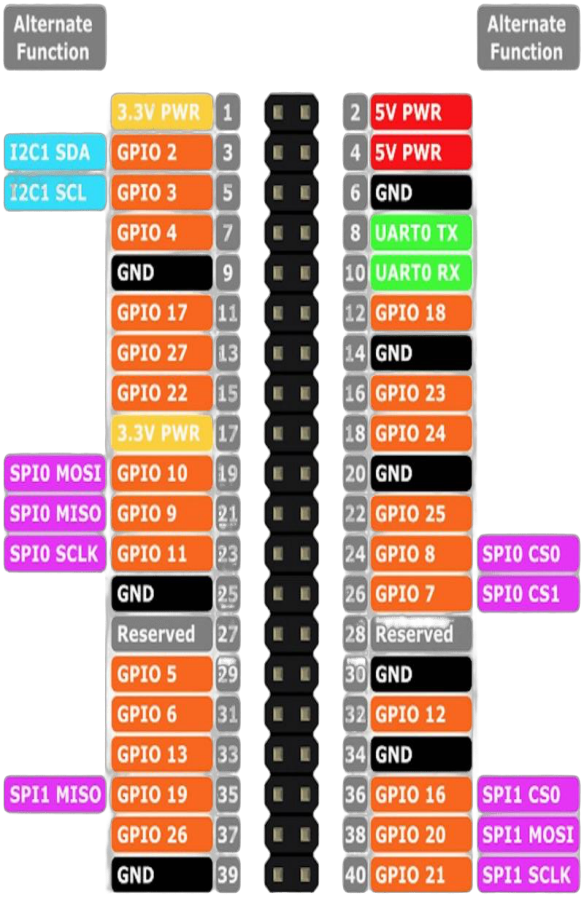
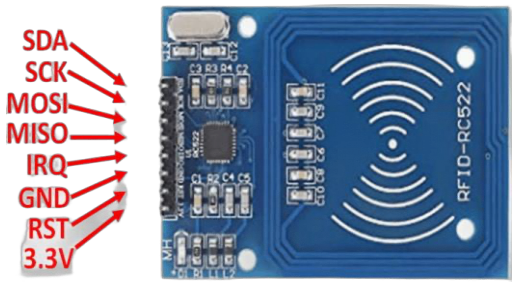
PRACTICAL 7 (Module 1)

TITLE : Interface RFID module with Raspberry Pi to read and write to RFID tags/cards

- **AIM :** To interface RFID module with Raspberry Pi to read and write to RFID tags/cards
- **HARDWARE COMPONENTS REQUIRED :** Raspberry Pi 400 (Pi version 4), power adaptor, HDMI to micro HDMI cable, mouse, monitor, RC522 RFID module, RFID card, jumper wires
- **CIRCUIT SCHEMATIC :**





NOTE : Draw and show the connections of the Raspberry Pi with the RFID module as per the reference table given below :

(Draw the exact same diagrams for the raspberry pi and the modules as the image attached aside with all the pin numbers and names)

RC522 module pins	Raspberry Pi pins	Raspberry Pi (Board/Physical pins)
3.3V	3.3V	1
Gnd	Gnd	6 (Can connect with any other groung pin)
SDA	SPI0 CS0	24
SCK	SPI0 SCLK	23
MOSI	SPI0 MOSI	19
MISO	SPI0 MISO	21

- **COMMANDS :**

  - sudo apt-get update (Updates the local list of available software packages)

  - sudo apt-get upgrade (Updates software packages installed on linux system)

  - sudo raspi-config (Enable SPI)

  - lsmod | grep spi (Checks, loads and runs the SPI drivers)

  - sudo apt -get install python3-dev python3-pip (Installs python 3 packages required)

  - sudo pip3 install spidev (Installs python 3 modules required to communicate with the SPI devices)

  - sudo pip3 install mfrc522 (Installs MFRC522 python library)

- **PROGRAM :**

  - **To write data :**

```
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522
r=SimpleMFRC522()
while True:
        text=input('Enter new data : ')
        print('Place the tag')
        r.write(text)
        print('Data written')
```

  - **To read data :**

```
import RPi.GPIO as GPIO
from mfrc522 import SimpleMFRC522
r=SimpleMFRC522()
while True:
        print('Place the tag')
        id,text=r.read()
        print('ID =',id)
        print('Data =',text)
```

NOTE : The program to write and read can be merged

- **RESULS & CONCLUSION : Note –** Write the output displayed on the Serial monitor (If no output got - Mention the errors displayed)